ATC 2.1.7 Release Notes

Advanced Trigonometry Calculator 2.1.7 release notes

Release date: 2026-06-09 Last updated: 2026-06-27

Highlights

- Added persistent switching between `double` and Boost `mp_float`.

- Fixed 50-decimal-place output for high precision constants such as `pi` and

e.

- Fixed `solve equation(x^7-12)` and other simple textual polynomial equations.

- Fixed `simplify polynomial(...)` for simple textual polynomials and simple

products of polynomial factors.

- Added an automatic regression suite based on the full user guide.

- Improved command-line automation for commands that export reports to `.txt`.

- Reduced RAM use and fixed several dynamic-allocation issues in sorting, DSP,

statistics, update checking, startup and date/time parsing code paths.

- Added a memory stress runner for repeated polynomial simplifier,

roots-to-polynomial and equation solver commands.

- Fixed direct `min(...)`, `max(...)` and `avg(...)` list calculations and

validated the same functions with matrix variable names.

- Relaxed user variable-name restrictions while preserving exact reserved

function, constant and numeric-prefix names.

- Added regression coverage for automatic multiplication deduction.

- Improved Windows 11 console behavior: the ATC intro is disabled by default on

Windows 11, console colors are mapped correctly when Windows Terminal uses ANSI/VT sequences, and ATC can open new instances as Windows Terminal tabs when `wt.exe` is available.

- Improved `verbose resolution`: direct `verbose resolution(1)` and

`verbose resolution(0)` commands are supported, and verbose output is no longer enabled during startup internals.

- Added exact rational-cancellation fast-path coverage for solver/equation

paths using real roots, `pi`, e and complex `pii` constants.

- Added interactive prompt improvements: Tab completion, repeated-Tab cycling

for ambiguous matches, broader command/function suggestions, dynamic user function suggestions and Up/Down history navigation.

- Rewrote and published the repository README to the GitHub `main` branch in

commit `f71e507 Improve project README`.

Documentation policy

Project documentation should be maintained in Portuguese and English from this point forward. Code comments, when present, should be written in English only.

Documentation improvements

ATC 2.1.7 documentation was expanded to make the project easier to learn, test and contribute to:

- added a one-page Quick Start near the beginning of the user guides;
- added an ATC Cookbook with practical workflows and safe example commands;
- added Best Practices for reliable use and reproducible bug reports;
- expanded the Architecture Overview with module and execution-flow diagrams;
- added a Developer Reference with contribution checklists;
- reorganized the full user guide around a more natural learning path;
- linked user documentation, coverage notes and regression status more clearly;
- generated bilingual Markdown/PDF documentation and DOCX full user guides.

Precision mode persistence

ATC now reads and writes:

```
%USERPROFILE%\Pictures\Advanced Trigonometry Calculator\higherPrecision.txt
```

Values:

- `0`: `double`
- `1`: Boost `mp_float`

Users can switch modes with:

```
higherprecision(1)
higherprecision(0)
```

The selected mode is applied after restarting ATC.

High precision fixes

The command:

```
dp50dppi
```

now returns:

```
3.14159265358979323846264338327950288419716939937511
```

The command:

```
dp50dpe
```

now returns:

```
2.71828182845904523536028747135266249775724709369996
```

Technical changes:

- fixed decimal output no longer uses `printf("%f")` with Boost multiprecision

values

- `pi` and e use Boost constants in `mp_float` mode
- decimal-place parsing avoids contaminating result state for simple integer

precision values

Equation solver fixes

ATC now normalizes simple textual polynomial equations before passing them to the coefficient-based solver.

Validated commands:

```
solve equation(x+2)
solve equation(x^2-5*x+6)
solve equation(x^7-12)
```

Expected examples:

```
solve equation(x+2)
x1=-2

solve equation(x^2-5*x+6)
x1=3
x2=2
```

Additional validated rational and symbolic-constant cases include:

```
solve equation(((x-5)(x+2))/(x-5))
solve equation((((x-5)(x+2))/(x-5))*(((x-5)(x+pi))/(x-5))*(((x-5)(x+e))/(x-5)))
solve equation(((x-e+pii)(x-e-pii))/(x-e-pii))
solver(((x-5)(x+2))/(x-5))
solver((x-e+pii)(x-e-pii))
```

For:

```
solve equation(x^7-12)
```

ATC now returns seven roots:

```
x1=1.42616
x2=0.889197+1.11502i
x3=0.889197-1.11502i
x4=-0.317351+1.3904i
x5=-0.317351-1.3904i
x6=-1.28493-0.618788i
x7=-1.28493+0.618788i
```

Export automation

Commands that ask whether to export a report can now be driven by redirected stdin in automated runs.

This supports test flows such as:

```
1
C:\path\to\report.txt
```

Interactive use remains unchanged.

Polynomial simplification fixes

`simplify polynomial(...)` now keeps `x` symbolic for simple textual polynomial inputs instead of evaluating it as zero before simplification.

Validated commands now include:

```
simplify polynomial(x)
simplify polynomial(x+1)
simplify polynomial(x+x+1)
simplify polynomial(x^2-5*x+6)
simplify polynomial((x-2)*(x-3))
simplify polynomial((x+1)*(x+1))
simplify polynomial((2*x+1)*(x-3))
```

Example:

```
simplify polynomial((x-2)*(x-3))
(1+0i)x^2+(_5+0i)x^1+(6+0i)
```

Exact cancellable rational products are also covered, including:

```
simplify polynomial(((x-5)(x+2))/(x-5))
simplify polynomial((((x-5)(x+2))/(x-5))*(((x-5)(x+2))/(x-5)))
```

Verbose resolution

`verbose resolution` remains available as an interactive menu. ATC 2.1.7 also supports:

```
verbose resolution(1)
verbose resolution(0)
```

The setting is persisted in `verboseResolution.txt`, but it is no longer applied globally during startup. This prevents verbose traces from internal startup work while preserving useful calculation traces for user expressions.

Verbose output was also made easier to read: display-only trailing `+0` terms are hidden, evaluated function arguments are labelled directly, function results are printed without the old interactor wording, and internal menu prompts keep their evaluation quiet even when verbose resolution is enabled.

Interactive prompt and autocomplete

The main ATC prompt now supports line editing and autocomplete while the user is writing an expression. Pressing `Tab` completes the closest documented command, mathematical function, alias or user function. When several matches are possible, repeated `Tab` presses cycle through them.

Autocomplete suggestions are ordered by shortest closest match first and then case-insensitively. The same expression can receive multiple completions, which is useful for longer mathematical input.

The prompt also supports `Up` and `Down` history navigation and normal cursor editing keys. The completion list is cached per input line to avoid rebuilding the vocabulary repeatedly while the user cycles suggestions.

Memory and allocation fixes

Several 2.1.7 code paths no longer reserve large global buffers when the input size is known:

- numeric sorting now allocates result buffers based on the number of submitted

values and frees them before returning

- `fft(...)` and `ifft(...)` allocate numeric buffers based on the number of

samples instead of `DIMDOUBLE`

- the even-size `fft(...)` loop no longer reads past the logical end of the

input and no longer depends on oversized zero-filled arrays

- statistics menu operations free temporary arrays and allocate list buffers

based on the number of submitted values

- dynamic string helpers now consistently preserve null termination and handle

null inputs more safely

- deletion helpers now release arrays directly instead of clearing every element

immediately before `delete[]`

- normal and long character-array deletion helpers now use the correct buffer

capacity when validating before release

- update checking now owns the HTTP response with `std::string` and closes

WinHTTP handles before returning

- startup reuses the global `forsprintf` buffer instead of leaking the first

allocation

- date/time command paths now use small stack buffers for two/four-character

fields instead of allocating `DIM`-sized temporaries

- alphabetical sorting normalization now avoids repeated `DIM` allocations for

one/two-character replacement keys

- the dynamic-allocation tracker is disabled by default so internal

`Variable-> ... variableData-> ...` diagnostics no longer appear in normal interactive output

- `function study(...)` now releases `poleZerosR` and `poleZerosI` before

returning, avoiding repeated memory growth in that path

- Release builds now set linker heap reserve/commit explicitly

(`x64: /HEAP:150000000,4194304`; `Win32: /HEAP:50000000,1048576`) and remove duplicate manual `/STACK` options from `AdditionalOptions`

Variable-name improvements

ATC now accepts broader Latin-letter variable names such as `data`, `energy`, `beta`, `alpha`, `matx`, `matrixvar`, `sinx` and `bvar`. This removes many legacy single-letter parser restrictions without allowing exact collisions with reserved names such as `sin`, `log`, `pi`, `e`, `i`, `B`, `O`, `H`, `P`, `D`, `b`, `min`, `max` and `avg`.

The parsing changes also protect guide syntaxes that use internal delimiter letters, including `rtD...D(...)` and `logb...b(...)`.

Automatic multiplication coverage

The suite now validates inferred multiplication across constants, functions, parentheses, imaginary values, custom guide syntax and variables. Covered forms include `2pi`, `pi2`, `2e`, `2sin(pi/2)`, `sin(pi/2)2`, `2(3+4)`, `(1+1)(2+3)`, `2i`, `3(2i)`, `2rtD4D(16)`, `2logb2b(8)`, `2x`, `x2`, `x(4)`, `(4)x`, `xy` and `x y`.

Composite variable names keep priority after whitespace normalization. If `xy` exists, `xy` and `x y` resolve to that variable; if only `x` and `y` exist, the same forms resolve as `x*y`.

Regression testing

New files:

```
tests\run-atc-regression.ps1
tests\run-atc-memory-stress.ps1
tests\ATC_AUTOMATED_TEST_CASES.md
```

Run:

```
powershell -ExecutionPolicy Bypass -File .\tests\run-atc-regression.ps1 -AtcExe .\x64\Release\atc.exe
```

Current result for both Release x64 and Release x86:

```
Summary: 374 passed, 0 failed
```

Current isolated coverage result:

```
Summary: 68 passed, 0 failed
```

Current SourceForge package validation result:

```
Summary: 44 passed, 0 failed
```

Memory stress run:

```
powershell -ExecutionPolicy Bypass -File .\tests\run-atc-memory-stress.ps1 -Iterations 10
```

Latest Release x64 10-iteration peak working-set observations:

```
simplify polynomial(...)       ~115-117 MB
roots to polynomial(...)       ~115 MB
solve equation(x^2-5*x+6)      ~141 MB
solve equation(1\8\1\_42)      ~144 MB
solve equation(x^7-12)         ~165 MB
```

Covered areas:

- arithmetic

- constants

- trigonometry and inverse trigonometry

- hyperbolic and inverse hyperbolic functions

- logarithms

- custom arithmetic/log guide syntax: `rest`, `quotient`, `rtD...D`,

`logb...b`, `afact`

- matrix/list guide syntax: `min`, `max`, `avg`, including matrix variables

- matrix guide helpers: `linsnum`, `colsnum`, `getlins`, `getcols`

- direct matrix/vector indexing and persisted indexed matrix updates

- relaxed variable names persisted through `variables.txt`

- automatic multiplication deduction
- digital signal processing: `sinc`, `fft`, `ifft`
- probability/statistics inline functions
- sorting with `.txt` report export
- ASCII and inverse ASCII sorting interactive flows
- roots-to-polynomial report export
- textual polynomial simplification
- exact rational polynomial cancellation in simplifier, solver and equation

solver paths

- quadratic equation solver
- 2x2 equations system solver
- textual polynomial equation solver
- coefficient-form cubic equation solver
- high-degree coefficient equations and internal expanded polynomial forms
- graph settings display, graph settings mutation, graph smoke tests,

deterministic graph navigation simulation, navigation edge clamping and explicit graph-range navigation

- TXT detector / command bridge commands, including `atc over cmd`, mocked

`atc from cmd`, mocked `to solve`, mocked `enable txt detector`, safe `eliminate strings`, and real-file `solve txt` response generation

- file/folder command existence checks
- safe app-environment runtime paths, including `about` and

`auto adjust window`

- safe runtime smoke coverage for deep interactive modules: unit conversions,

microeconomics, physics, statistics, geometry, financial calculations, triangles rectangles solver and arithmetic matrix solver

- persistent settings and interactive menu retry behavior
- interactive prompt autocomplete vocabulary, repeated-Tab cycling and

Up/Down history navigation

- long-running time command branches, short stopwatch marking, zero-duration

clock checks and timer syntax guards

- verbose-resolution persisted setting, cleaned verbose output behavior, and

suppression of verbose traces for internal menu inputs

- Windows 11 console source-level behavior checks
- `double` and Boost `mp_float` precision behavior
- restart persistence of precision mode

Build verification

Release x64 build succeeds with MSBuild:

```
& 'C:\Program Files\Microsoft Visual Studio\18\Insiders\MSBuild\Current\Bin\MSBuild.exe' '.\Advanced Trigonometry C
```

Release Win32/x86 builds are also supported by the solution and remain a compatibility target for older Windows versions. The current regression suite has been validated against both `x64\Release\atc.exe` and `Release\atc.exe`.

The latest Release x86 and Release x64 builds completed successfully with `0 Warning(s), 0 Error(s)`.

Known limitations

The automatic suite intentionally avoids:

- shutdown, restart, hibernate, sleep and lock commands
- full-duration clock, stopwatch and timer views
- commands that open external browsers or editors, except for TXT/file flows

covered through explicit test mocks

- full live manual graph rendering and live key-buffer behavior
- remaining branches inside deeply interactive modules such as triangles,

arithmetic matrix solver, conversions, finance, physics, microeconomics and statistics menus

Files touched in this release work

Important code areas:

```
Advanced Trigonometry Calculator\main.cpp
Advanced Trigonometry Calculator\auto_complete.cpp
Advanced Trigonometry Calculator\commands.cpp
Advanced Trigonometry Calculator\dirent.h
Advanced Trigonometry Calculator\main_aux_processor.cpp
Advanced Trigonometry Calculator\processing_core.cpp
Advanced Trigonometry Calculator\scripting.cpp
Advanced Trigonometry Calculator\stdafx.h
Advanced Trigonometry Calculator\dynamic_allocations.cpp
Advanced Trigonometry Calculator\digital_signal_processing.cpp
Advanced Trigonometry Calculator\sorting.cpp
Advanced Trigonometry Calculator\statistics_calculations.cpp
Advanced Trigonometry Calculator\check_for_updates.cpp
Advanced Trigonometry Calculator\versioninfo.rc
README.md
```

Documentation and tests:

```
docs\ATC_2.1.7_DOCUMENTATION.md
docs\RELEASE_2.1.7.md
tests\ATC_AUTOMATED_TEST_CASES.md
tests\ATC_USER_GUIDE_COVERAGE.md
tests\run-atc-regression.ps1
tests\run-atc-isolated-coverage.ps1
```

Resumo de release em portugues

Esta release 2.1.7 reforca a estabilidade tecnica do ATC e prepara o projeto para uma distribuicao mais clara como projeto open-source.

Principais pontos em portugues:

- alternancia persistente entre `double` e Boost `mp_float`;
- correcao da formatacao de alta precisao, incluindo `dp50dppi` e `dp50dpe`;
- melhorias em `solve equation`, `solver` e `simplify polynomial`, incluindo

produtos racionais com fatores cancelaveis e constantes simbolicas como `pi`, e e `pii`;

- melhoria de `verbose resolution`, agora com `verbose resolution(1)` e

`verbose resolution(0)`, evitando logs internos durante o arranque;

- melhorias no comportamento da consola no Windows 11 e suporte para novas

instancias/abas quando o Windows Terminal esta disponivel;

- reducao de alocacoes desnecessarias e melhoria de libertacao de memoria em

varios caminhos;

- suite automatizada atual com `359 passed, 0 failed` em Release x64 e

Release x86;

- cobertura isolada atual com `65 passed, 0 failed`, incluindo autocomplete e navegacao por historico;
- README publico reescrito e publicado no GitHub em `main` no commit `f71e507 Improve project README`;
- a documentacao futura deve ser mantida em portugues e ingles, enquanto os comentarios no codigo devem permanecer apenas em ingles.